SMART INDIA HACKATHON 2026

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- Problem Statement ID - SIH26249
- Problem Statement Title - Air Power - Predictive Maintenance & Fleet Availability
- Theme - Transportation & Logistics
- PS Category - Software
- Team ID - 137722
- Team Name (Registered on portal) - Backpropagators



Backpropagators

FleetSense

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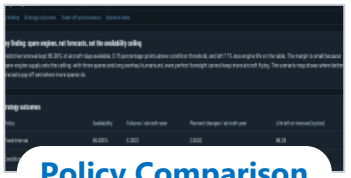
Proposed Solution

- An **on-premise readiness console** joining engine health data, maintenance records and **spares stock** per airframe.
- **Calibrated failure risk** for every engine at **8 horizons** (10–100 cycles).
- **Risk-based removal planner** that books engine changes against **slots, crews and spare engines**.
- **Policy lab** tests fixed-interval, condition-based and predictive rules on a **simulated squadron-year** before adoption.
- **Independent rule checker** re-verifies **every schedule** the planner produces.
- Runs **air-gapped** on a standard CPU server, built to read **e-MMS and IMMOLS** exports.



Fleet Replay

every airframe, its engine and 20-cycle risk



Policy Comparison

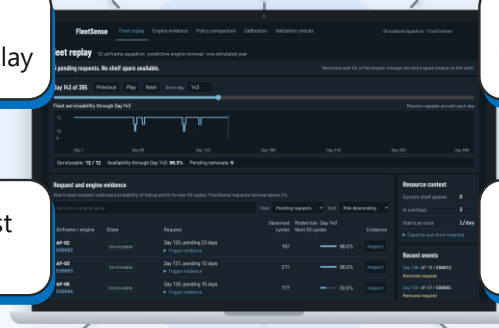
test maintenance rules before adopting them

air-gapped, no data leaves base

calibrated failure risk per engine

spares-aware removal planner

year-long readiness replay



engine trend vs fleet norm

policy lab: test rules first

independent rule checker

warnings before failure

scales to 200+ airframes

e-MMS and IMMOLS ready

"every airframe ready, every engine accounted for"

It Addresses the Problem by

- **Joining fragmented data:** sensors, logs and spares in **one record per engine**.
- **Moving from reactive to risk-based** removal with calibrated probabilities.
- **Protecting availability:** removals planned against **real slot and spare limits**.
- **Building trust:** every model tested on **unseen engines**, every plan re-checked.

Base server | Air-gapped LAN | Any browser | Docker



FleetSense console

- Engineering officer: daily removal list
- Logistics: spares and overhaul queue
- Commander: mission-capable count
- Air HQ: policy before rollout

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TECHNICAL APPROACH

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Tech Stack

Machine learning

Python 3.12 | LightGBM | scikit-learn | NumPy | pandas

Planning engine

Squadron simulator | Removal planner | Rule checker

Backend

FastAPI | Uvicorn | Parquet

Frontend

HTML/CSS/JS | Plotly.js

Data

NASA C-MAPSS | e-MMS / IMMOLS exports

Testing & deploy

pytest (63 tests) | Playwright | Docker

Fully open-source stack that runs offline on base servers

Our Working Prototype: click here

end-to-end: sensor data → risk → plan → console

engine sensor + records → LightGBM risk models → isotonic calibration → removal planner → rule checker

fleet board
engine health
removal list

engineering officer

spares on shelf
overhaul queue
spares forecast

logistics officer

readiness replay
policy lab
system checks

commander / Air HQ

Base network (air-gapped): loopback and LAN only, no outbound connections

Browser console
fleet · engines · policy lab

FastAPI server

Risk model
LightGBM · 8 horizons · isotonic calibration

Removal planner
slots, crews and spare engines

Rule checker
re-verifies every schedule independently

Engine health data
HUMS / sensor exports

Maintenance records
e-MMS export, read-only

Spares stock
IMMOLS export, read-only

Local store
Parquet, JSON, audit log

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FEASIBILITY AND VIABILITY

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Financials

iDEX grant
MoD innovation funding

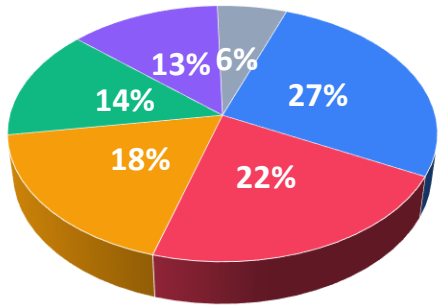
Pilot squadron
12-month field trial

Command rollout
no licence fees

Annual support
model upkeep

Estimated pilot budget · one squadron, 12 months

- **Data integration** ₹6L - ₹8L e-MMS, IMMOLS, sensors
- **Model and validation** ₹5L - ₹6L training, calibration
- **Console and training** ₹4L - ₹5L UI, user training
- **On-prem hardware** ₹3L - ₹4L CPU server, storage
- **Security audit** ₹3L - ₹4L accreditation, testing
- **Contingency** ₹1.5L - ₹2L reserve



Feasibility

Working prototype: sensor data to removal plan runs **end-to-end on a laptop CPU**.

Proven open tools: LightGBM, scikit-learn, FastAPI. **No licence cost**, no internet.

Tested on unseen engines: failures flagged in advance **86.5%** of the time.

Scales: 200 airframes, a full year simulated and re-checked in **0.64 s**.

Viability

IAF fields **31 fighter squadrons against 42 sanctioned**: every available airframe counts.

CAG found Su-30MKI serviceability at **55–60% vs a 75% norm**, driven by spares and AOG.

Predictive maintenance cuts **breakdowns 70–75%** and **downtime 35–45%** (US DOE).

India's MRO market: **~$2B (2024) to $4B by 2031**, an indigenous capability.

Challenges and Mitigation

Classified, siloed data	**On-premise, air-gapped** deployment with **read-only adapters**; no data leaves the base.
Limited failure history for some fleets	Learn from **public run-to-failure data**, then **calibrate on each fleet's** own engines.
Risk scores **ignored or over-trusted**	**Calibrated probabilities** with fleet-norm evidence; an **engineer approves** every removal.
Spares, not prediction, limit readiness	**Policy lab** shows when **more spares beat better forecasts**, before money is spent.

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IMPACT AND BENEFITS

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- **Higher availability** from the same fleet: engines pulled **before** they fail.
- **86.5% of failures flagged in advance** on engines the model never saw.



- **Engineering officers:** one prioritised removal list instead of scattered logbooks.
- **Technicians:** sensor trends against fleet norm for every engine.

- **11% less engine life wasted** than condition-based removal in our squadron simulation.
- Spares demand **forecast weeks ahead** from predicted removals.

- **Logistics:** spares and overhaul queue visible every day.
- **Commanders:** mission-capable count and year-long readiness at a glance.

- Runs **air-gapped**: no maintenance data leaves the base.
- **Zero licence cost**: open-source stack on existing servers.

IMPACT



BENEFITS

- **Air HQ / MoD:** evaluate maintenance policy before fleet-wide rollout.
- **HAL and BRD depots:** steadier overhaul demand from planned removals.



- Every maintenance rule **tested on a simulated squadron-year** before adoption.
- **0 rule violations** across **44,820** checked schedules.

- **Beyond IAF:** Army Aviation, Navy, civil MRO and rail rolling stock.
- **Atmanirbhar:** indigenous, open and sovereign maintenance intelligence.



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RESEARCH AND REFERENCES

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Research Papers

Prognostics and maintenance

- Damage propagation modeling for aircraft engine run-to-failure simulation - Saxena et al., IEEE PHM, 2008 [Link]
- Predictive maintenance for multi-component systems of repairables with RUL prognostics and limited spares - de Pater & Mitici, RESS, 2021 [Link]

Machine learning

- LightGBM: A Highly Efficient Gradient Boosting Decision Tree - Ke et al., NeurIPS, 2017 [Link]
- Probability calibration (isotonic regression) - scikit-learn documentation [Link]

Defence and industry

- CAG audit of Su-30MKI serviceability - Defense News, 2015 [Link]
- Parliamentary panel: IAF at 31 squadrons vs 42 - Deccan Herald, 2024 [Link]
- Benefits of predictive maintenance (US DOE O&M Guide) - HFM Magazine [Link]

Our Demo Video: click here

3-minute walkthrough of the working prototype: fleet board, policy lab and model trust

Our Prototype: click here

Competitors Comparison

	FleetSense	Manual logs	Generic CMMS	Cloud PdM
Failure-risk forecasting	✓	✗	✗	✓
Runs air-gapped on base	✓	✓	✓	✗
Spares-aware planning	✓	✗	✗	✓
No per-seat licence	✓	✓	✗	✗
Open, auditable models	✓	✗	✗	✗

Industry Outlook

31 / 42	IAF fighter squadrons in service vs sanctioned (2024)
55–60%	Su-30MKI serviceability vs a 75% norm (CAG)
70–75%	fewer breakdowns with predictive maintenance (US DOE)
$4B	India MRO market by 2031, from ~$2B in 2024

Prototype Benchmarks

86.5%	of failures on unseen engines flagged in advance (609 of 704)
0	maintenance-rule violations across 44,820 simulated squadron-years
0.64 s	to simulate and re-check 200 airframes for a full year
63	automated tests across model, planner, checker and console